

Nallabati Venkata Durga Sai Sarat Chandra

Target Role: Data Engineer Intern | PhonePe

saratchandranallabati@gmail.com | +91 6305761091 | Kollam | [Research Gate](#) | [Linkedin](#) | [Github](#) | [Portfolio](#)

SUMMARY

Visionary Robotics & AI undergraduate specializing in the intersection of edge mechatronics, Scientific Machine Learning, and human-centric design. Combines hands-on development of autonomous hardware and advanced physics-informed models with intuitive UI/UX architecture. Poised to deliver high-impact, user-friendly edge intelligence systems that seamlessly bridge complex physical dynamics with operational clarity.

EDUCATION

Bachelor of Technology (B.Tech), Amrita Vishwa Vidyapeetham

CGPA: 9.05

2024 – Present

Amritapuri

Higher Secondary Education, Sri Chaitanya Vidhyanikethan

2024

CBSE (MPCEA) Percentage: 88.2%

Vishakapatnam

Secondary School Education, Sri Chaitanya School

2022

SSC AP Marks: 570/600

Rajam

SKILLS

Core Stack & Systems: Python, C, C++, SQL, Git, Linux, Ubuntu Server, ROS 2

AI, ML & SciML: Physics-Informed Machine Learning (PIML), Deep Learning, Optuna (Bayesian Optimization), Explainable AI

Data Analytics & Modeling: Pandas, NumPy, Scikit-Learn, LightGBM, XGBoost, Conformal Prediction, Time-Series DSP

Engineering & Simulation: Computational Fluid Dynamics (ANSYS Fluent), Thermodynamics, CAD (Fusion 360, AutoCAD)

Design & UI/UX Tools: Figma, Adobe Photoshop, Canva, UI/UX Wireframing, Responsive Layouts

EXPERIENCE

InAmigos Foundation (Internship), *Graphic Designer*

06/2025

- Developing visual content, including posters and reels, for various organizational programs to support fundraising initiatives and enhance public engagement across social media platforms. Utilized Adobe Photoshop, Canva and some AI tools to produce engaging visual assets.

PROJECTS

Domain-Aware Industrial Failure Prediction & Diagnostics Framework

- Engineered and optimized system architecture, telemetry protocols, and validated performance under operational constraints.

Blood Neural Network: PINNs for Inverse Hemodynamic Rheology Discovery

- Engineered and optimized system architecture, telemetry protocols, and validated performance under operational constraints.

Multi-Parameter Health Vitals Monitor & Real-Time IoT Data Streamer

- Engineered and optimized system architecture, telemetry protocols, and validated performance under operational constraints.

Evolution Edge: Self-Evolving Heterogeneous Neural Bridge

- Engineered and optimized system architecture, telemetry protocols, and validated performance under operational constraints.

Dynamic Diffusion: Physics-Guided 3D Molecular Generation

- Engineered and optimized system architecture, telemetry protocols, and validated performance under operational constraints.

PUBLICATIONS

Small Scale Raspberry Pi Based Autonomous Car: A Practical Guide to Understand Complexity, *IEEE ICRM 2025*

11/2025

- Architecting Level 4 autonomous navigation logic within a 2.4 GHz CPU hardware bottleneck, utilizing Probabilistic Hough Transforms for real-time computer vision processing, and engineering a handshake-free UDP telemetry protocol to eliminate multi-agent V2V latency in edge environments.

CLUB EXPERIENCE

Embedded Systems and Robotics Developer, *CLUB MIRAI*

12/2025 – Present

- Utilising ROS 2 Jazzy to create a low-latency distributed control system for actuator and perception nodes. Developing a 2D/3D kinematic model for a companion robot and applying inverse kinematics (IK) for motor joint angles. Implementing optimised libraries for high frame-rate detection in a dynamic face tracking computer vision pipeline. Establishing a Raspberry Pi 5 remote development environment on Ubuntu Server for HIL testing and headless ROS 2 execution.

ACHIEVEMENTS

- Selected among top international teams to complete all three technical milestones of the IEEE IES Generative AI Challenge.
- Successfully cleared the 2 Levels of NAEST (India's premier physics experimental competition organized by IAPT and Dr. H.C. Verma).
- Selected for the ISRO-IIRS YUVIKA through a national-level merit screening and science aptitude evaluation.
- Qualified for MTSO (Maths Olympiad) Level 2 (Stage II) by ranking in the top tier of candidates nationwide.